PROJECT WORK – III
SYNOPSIS REPORT

A Lightweight LLM + RAG + CAG Framework for Medical Health Question Answering

by

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Entry No.: 23BCS040

Under the Guidance

Of

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1. Title of the Project

A Lightweight LLM + RAG + CAG Framework for Medical Health Question Answering (prototype name: *MedRAC-QA*, a Retrieval- and Cache-Augmented small language model for evidence-grounded medical question answering).

2. Name of the Student with Entry Number

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4. Introduction and Background of the Study

People increasingly ask health questions to chatbots built on large language models (LLMs). These models write fluent answers, but they can *hallucinate*: state plausible facts that are wrong or that no trusted source supports. In health, an unsupported answer can cause real harm. Large models are also expensive to run, and the same common questions ("What are the symptoms of anaemia?") are asked again and again in different words.

Two techniques address these problems. **Retrieval-Augmented Generation (RAG)** first retrieves passages from a trusted knowledge base and asks the model to answer only from them, with citations. **Cache-Augmented Generation (CAG)** reuses knowledge or answers that were prepared earlier instead of recomputing them, which cuts latency and cost. A careless cache is dangerous in medicine, however. "Symptoms of diabetes" and "symptoms of diabetes *in children*" look almost identical to an embedding model but need different answers.

This project builds a **lightweight** framework that combines a small, fast LLM with RAG over NIH health content (MedQuAD) and a **safety-constrained semantic cache**. A cached answer is reused only when the new question matches it in meaning, intent, clinical context, topic and evidence freshness. The system is evaluated on public medical QA benchmarks for quality, efficiency and safety.

5. Objectives of the Project

1. Develop a lightweight, evidence-grounded medical QA framework that uses a frozen compact instruction-tuned LLM with retrieval over a curated, licensed medical knowledge base (MedQuAD).
2. Design a **semantic answer cache (CAG)** whose admission and reuse are gated by question similarity, clinical-context compatibility (intent, population, negation, type numbers, topic), source freshness and evidence validity.
3. Implement an **evidence-sufficiency gate** and **sentence-level citation validation**, so that the system abstains or shows sources instead of giving unsupported answers.
4. Implement rule-based **safety screening** for emergencies (directing users to 112 / 108 / Tele-MANAS 14416) and for questions that must never be answered from a cache (doses, personal details, time-sensitive questions).
5. Evaluate the trade-off between answer quality, retrieval quality, cache correctness, latency and cost against baselines (LLM-only, RAG-only, exact-match cache, plain semantic cache), with ablation studies.
6. Deliver a deployable web application with an API that shows every answer's citations and explains whether it came from the validated cache or from retrieval.

6. Problem Statement

General-purpose LLMs can produce medically inaccurate answers, often do not reflect the evidence in a supplied knowledge base, and repeatedly spend retrieval and inference cost on questions that have effectively been answered before. Conventional RAG improves grounding but still retrieves and generates for every query, while naive caching can return an answer written for a clinically different question (a different age group, pregnancy status or disease subtype).

Problem. Given a health question q, a trusted corpus D and a cache C of previously validated answers, produce a concise answer $\hat{y} = f(q, D, C)$ with supporting citations, or a safe abstention, such that (a) every claim is supported by D, (b) cached answers are reused only when they are clinically appropriate for q, and (c) average latency and cost fall without loss of answer quality.

Research questions. RQ1: Does RAG improve the factual grounding of a compact LLM over direct generation? RQ2: Does semantic CAG reduce latency and cost without degrading quality? RQ3: Does context-aware cache validation reduce incorrect reuse for clinically dissimilar questions? RQ4: Do evidence gating and citation validation reduce unsupported answers? RQ5: How do the retrieval depth k and the thresholds τ (cache) and γ (evidence) affect performance?

7. Methodology / Approach

Knowledge base. MedQuAD (47,457 QA pairs from 12 NIH websites, CC BY 4.0) is parsed; the three subsets whose answers were withdrawn for copyright reasons are excluded. Answers are de-duplicated and split into overlapping 180-word chunks (fitting the 256-token window of the encoder). Each chunk, prefixed with its source question, is embedded with *all-MiniLM-L6-v2* (run through ONNX Runtime on CPU) and L2-normalised, so cosine similarity $s(q, p) = \hat{q}^T\hat{p}$ is a dot product; the top-k passages are retrieved by exact search.

Pipeline. (1) Rule-based safety screening; emergencies receive fixed guidance with no model involved. (2) The question is embedded once. (3) **CAG:** the most similar cached question c *is reused only if cos(q, c)* $\geq \tau$, the entry is fresh (same knowledge-base version, within its TTL), the two questions agree in intent, clinical qualifiers, type numbers and topic words, and all of its cited passages still exist. (4) Otherwise **RAG** retrieves k = 5 passages. (5) If the best score is below γ, the system abstains. (6) A lightweight LLM (Gemini Flash-Lite through an OpenAI-compatible API, interchangeable with a local Qwen2.5-3B via Ollama) answers only from the passages, citing them as [n]. (7) Each answer sentence is checked against its cited passages (semantic similarity $\geq \delta$ or high content-word overlap). Answers that fail fall back to showing the sources. (8) Validated, cacheable answers are admitted to the cache. The cache is shared by all server instances through a serverless Redis store, so a verified answer produced once can be reused by every user, and survives restarts.

Evaluation. Baselines: B1 LLM-only, B2 RAG, B3 exact-match cache, B4 plain semantic cache, B5 RAG + semantic cache, B6 full MedRAC-QA. Datasets: MedQuAD (corpus and cache workload), TREC 2017 LiveQA-Med real consumer questions with expert-judged answers (retrieval and abstention), and PubMedQA (yes/no/maybe reasoning). Metrics: Hit@k, Recall@k and MRR for retrieval; cache useful-hit rate, hit precision and unsafe-reuse rate; abstention coverage and out-of-scope rejection; ROUGE-L, semantic similarity and supported-sentence rate for answers; median and p95 latency. Thresholds τ and γ are tuned on development splits only, ablations remove one component at a time, and paired bootstrap / McNemar tests assess significance.

8. Tools, Technologies, or Datasets to be Used

Component	Choice
Language	Python 3.12

Lightweight LLM	Gemini 3.5 Flash-Lite (free tier) via OpenAI-compatible API; Qwen2.5-3B-Instruct via Ollama as a local alternative
Embeddings	sentence-transformers/all-MiniLM-L6-v2 (ONNX Runtime, quantised, CPU)
Retrieval / cache	NumPy exact cosine search; custom semantic cache with a shared Upstash Redis store (in-memory / JSON file for local runs)
Web API and UI	FastAPI; HTML, CSS and JavaScript front end; deployed on Vercel (https://med-rac.vercel.app)
Testing	pytest (unit and API tests with test doubles), GitHub Actions CI
Datasets	MedQuAD (NIH, CC BY 4.0); TREC 2017 LiveQA-Med test set; PubMedQA (PQA-L)
Evaluation	Hit@k, Recall@k, MRR, cache precision / unsafe-reuse rate, ROUGE-L, latency; paired bootstrap and McNemar tests

9. Expected Outcomes / Deliverables

- A working, deployable medical QA web application that answers with citations to NIH sources, abstains when evidence is insufficient and gives emergency guidance when needed.
- A safety-constrained semantic cache (CAG) that serves repeated and reworded questions in milliseconds while refusing clinically inappropriate reuse.
- A measured comparison of the proposed system with LLM-only, RAG-only and naive-cache baselines on answer quality, retrieval quality, cache correctness and latency, with ablation studies.
- A reproducible codebase (data scripts, index builder, experiment scripts, tests, CI) and the project report.

10. Timeline of Work

Phase	Dates	Work
Phase 0 — Study and design	Early Oct 2026	Literature study (medical RAG, CAG, semantic caching), problem formulation, dataset selection
Phase 1 — Core framework	Oct 2026	Knowledge base and index, safety screening, semantic cache with context gate, retrieval, LLM generation, citation validation, web app, tests; cache, retrieval and abstention experiments
Mid-semester evaluation	Oct 2026	Deployed working demonstration and preliminary results (cache safety, retrieval, abstention, end-to-end pilot)
Phase 2 — Full evaluation	23 Oct – 8 Nov 2026	B1–B6 comparison, PubMedQA, hybrid retrieval and biomedical-encoder comparison, NLI-based claim checking, safety test set
Phase 3 — Completion	9 – 15 Nov 2026	Remaining ablations, knowledge-preloading CAG experiment, significance tests, final deployment
Report	16 – 22 Nov 2026	Project report, paper draft and demo video
Final evaluation	23 – 27 Nov 2026	Report, demonstration and viva

11. References

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